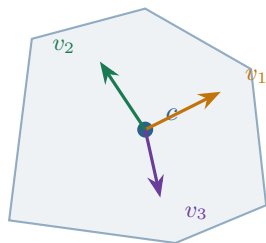


The Shape of Uncertainty

Star Set

(Feedforward NN Input)



Each generator v_i is a vector

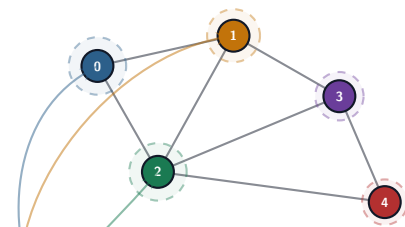
$$v_1 = \begin{bmatrix} 0.3 \\ 1.2 \\ -0.4 \\ 0.8 \end{bmatrix} \quad \begin{matrix} \updownarrow \\ n \end{matrix}$$

n = input dimension

$$S = \{c + \sum \alpha_i v_i \mid C\alpha \leq d\}$$

GraphStar Set

(GNN Input)



Each generator V_i is an $N \times F$ matrix

V_1
each row = a node

	f_1	f_2	f_3	f_4
0	0.2	-0.1	0.5	-0.3
1	0.2	-0.1	0.5	-0.3
2	0.2	-0.1	0.5	-0.3
3	0.2	-0.1	0.5	-0.3
4	0.2	-0.1	0.5	-0.3

N

F features

$$S = \{C + \sum \alpha_i V_i \mid C\alpha \leq d\}$$

Star generators are vectors (one value per input dimension) $\longleftrightarrow$ Graph-Star generators are matrices (one row per node, one column per feature)